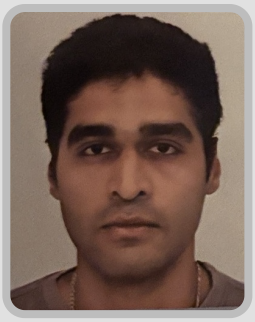
VINEETH RAGHAV GURURAJAN

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ABOUT ME

I am an Electrical Engineering student from India, now doing my Master's in Stuttgart (Electrical Power Systems) after a B.Tech. in Chennai. I enjoy electronics and simulation, from FPGA design in Verilog to battery models, and I like teaching what I learn: I have run electronics and coding workshops for beginners. I work carefully, document my results, and keep learning, including German (A2, preparing for B1). I am looking for a working student role in hardware, simulation or energy systems, about 20 hours per week.

EDUCATION

10/2025 – 09/2027 (expected) **M.Sc. Electrical Engineering, University of Stuttgart**
Specialisation Electrical Power Systems: Intelligent Power Grids, Power Systems Engineering, Battery Modeling

08/2021 – 05/2025 **B.Tech. Electronics and Computer Engineering, SRM Institute of Science and Technology, Chennai**
Grades A+ or A in FPGA-Based Embedded Systems, Embedded Hardware and Operating Systems, Microcontrollers and Interfacing, Hardware Interfacing and Networking, IoT System Design, Deep Learning; final-year project graded A+ (10 credits). Seminar I (11/2023, grade O): cryptocurrency price analysis and prediction, review of forecasting methods and evaluation metrics (MAE, RMSE).

PROJECTS

12/2025 **Battery Cell Modeling, team project, University of Stuttgart**

- Analysed cell-to-cell variance caused by electrical, thermal, aging and manufacturing tolerances by simulation.
- Implemented and debugged statistical parameter sampling in the simulation model.

01/2025 – 05/2025 **Optimized 64-bit RoBA Multiplier with MAC Unit, B.Tech. final-year project**

- Designed a rounding-based approximate multiplier with a multiply-accumulate unit in Verilog; shift-and-add replaces partial-product computation.
- Synthesised in Xilinx ISE (Spartan-3 target); reported critical path 0.575 ns.

11/2024 **Duplicate Question Detection, team project (4 students)**

- Classified duplicate question pairs from the Quora Question Pairs dataset using NLP and fuzzy-ratio features.
- SVM reached 88.3% and Random Forest 80.02% accuracy; deployed as a web application on Heroku.

SKILLS

Electronics and embedded: microcontrollers (ARM, ESP32, Arduino), FPGA design (Verilog, Xilinx ISE), PCB design (Altium), circuit simulation (PSpice), CAN, I2C, SPI, UART

Software and modeling: Python, C/C++, MATLAB, machine learning, statistical parameter sampling

Power and control: power electronics, control systems, sensors and actuators, battery modeling

LANGUAGES

English: C1 (IELTS Academic, overall band 7.5, October 2024) **German:** A2, preparing for the telc B1 exam
Kannada: native

CERTIFICATES AND VOLUNTEERING

PSpice Fundamentals (TVS India); Certificate of Training (Livewire Ltd.). STEM workshops in electronics and coding for underprivileged students; post-cyclone relief work in Chennai.